# Stat 415 Lab 2 - Recommender Systems Pt. 1

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## 1 Introduction

Our goal is to create a series of recommender systems for restaurants in Evanston, IL using past reviews of said restaurants. This will prove helpful for locals, newcomers, and Northwestern University students alike, as they decide which restaurants they want to try.

## 2 EDA

### 2.1 The Data

The raw data was collected from unspecified websites and contain 1500 reviews of real restaurants in Evanston, IL that people have written in the past. We used the variables:

- Reviewer Name — Name of the person writing the review.
- Restaurant Name — Name of the restaurant being reviewed.
- Rating — Numerical rating score given by the reviewer.
- Review Text — Textual content of the review.
- Date of Review — Date when the review was posted.
- Birth Year — Year of birth of the reviewer.
- Marital Status — Marital status of the reviewer (e.g., Single, Married).
- Has Children? — Whether the reviewer has children (Yes/No/Unknown).
- Vegetarian? — Whether the reviewer is a vegetarian (Yes/No/Unknown).
- Weight (lb) — Reported body weight of the reviewer in pounds.
- Height (cm) — Reported height of the reviewer in centimeters.
- Average Amount Spent — Typical spending amount category (e.g., Low, Medium, High).
- Preferred Mode of Transport — Usual transportation method of the reviewer.
- Northwestern Student? — Whether the reviewer is a Northwestern University student.

We initially find that columns *Review Text, Birth Year, Martial Status, Has Children?, Vegetarian? Height (lb), Weight(cm), Average Amount Spent,* and *Preferred Mode of Transport* have missing values. We find that a whopping 1405/1500 of the data points don't specify whether they're vegetarian, so we impute these missing values with "Unknown". We don't do mode imputation, because it would be catastrophic for our recommender system to suggest non-vegetarian restaurants to specifically vegetarian customers. The variable with the second most missing data points is *Review Text*, for which we just imputed the missing values with an empty string, since this variable is crucial for building our recommender system, but we don't want to add any extra words to skew the TF-IDF scores later on for

certain words (such as "No"). For the rest of the numerical variables (*Birth Year*, *Weight (lb)*, and *Height(cm)*, we performed median imputation (median instead of mean is safer because of outliers) and created a missing-flag column for each of them, since the missingness of the variables can have some implications for the kinds of people the reviewers are. For the rest of the categorical variables: *Martial Status, Has Children?,* and *Preferred Mode of Transport*, we impute with "Unknown" just as we did for the *Vegetarian?* variable. Lastly, we just did mode imputation for the categorical variable *Average Amount Spent* since only one data point was missing, and added a missing flag variable for this as well.

After all this, we added the new indicator variables:

- Birth Year_missing — Indicator flag for missing birth year data (1 if missing, 0 if present).
- Weight (lb)_missing — Indicator flag for missing weight data (1 if missing, 0 if present).
- Height (cm)_missing — Indicator flag for missing height data (1 if missing, 0 if present).
- Average Amount Spent_missing — Indicator flag for missing spending amount data (1 if missing, 0 if present).

## 2.2 Univariate Data Analysis

We see that the *Vegetarian?* variable is highly unbalanced with the majority of data points being missing, so we remove this. We can see the distribution of some of the of the categorical variables in 1.

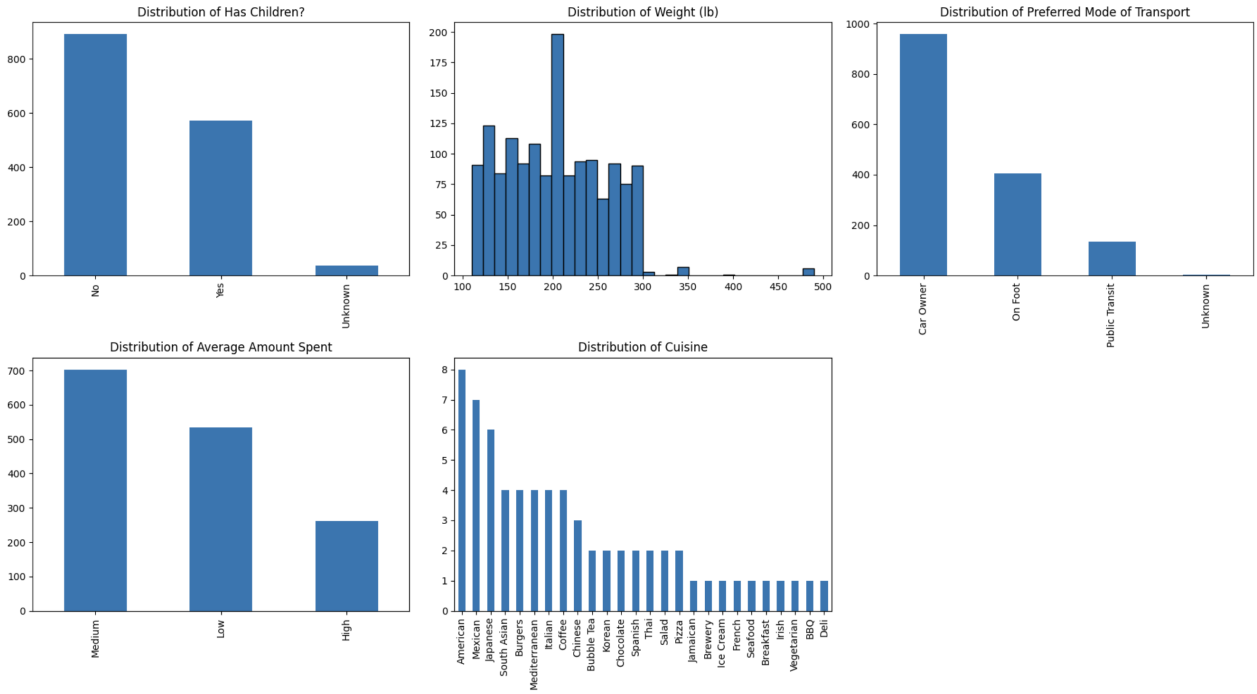



Figure 1: Bar Graphs of Five Categorical Variables

## 2.3 Grouping the Data

We see below that the average ratings per cluster, using three separate clustering techniques: K-Means, DBSCAN, and Agglomerative Clustering. We find that when we used K-Means clustering and when we used Agglomerative Clustering, all four clusters had similar average ratings (of around 4), which suggests that reviewers of different demographics have relatively similar ratings. On the other hand, when we used DBSCAN, there were significantly more clusters, with there being a greater variance in ratings. This makes sense to us, because creating smaller groups will lead to outliers more prominently affecting the mean of each group.

| | KMeans_cluster | n_users | avg_rating |
|---|---|---|---|
| 0 | 0 | 158 | 4.018987 |
| 1 | 1 | 540 | 3.592593 |
| 2 | 2 | 411 | 3.807786 |
| 3 | 3 | 391 | 3.780051 |

Figure 2: K-Means Clustering Summary

| | Agglo_cluster | n_users | avg_rating |
|---|---|---|---|
| 0 | 0 | 627 | 3.558214 |
| 1 | 1 | 321 | 4.049844 |
| 2 | 2 | 235 | 3.923404 |
| 3 | 3 | 317 | 3.675079 |

Figure 3: Agglomerative Clustering Summary

| | DBSCAN_cluster | n_users | avg_rating |
|---|---|---|---|
| 0 | -1 | 176 | 3.823864 |
| 1 | 0 | 133 | 3.992481 |
| 2 | 1 | 82 | 4.000000 |
| 3 | 2 | 22 | 3.409091 |
| 4 | 3 | 63 | 3.841270 |
| 5 | 4 | 22 | 2.863636 |
| 6 | 5 | 67 | 4.000000 |
| 7 | 6 | 103 | 3.368932 |
| 8 | 7 | 10 | 3.900000 |
| 9 | 8 | 154 | 3.948052 |
| 10 | 9 | 17 | 4.294118 |
| 11 | 10 | 20 | 3.350000 |
| 12 | 11 | 59 | 3.508475 |
| 13 | 12 | 40 | 3.625000 |
| 14 | 13 | 54 | 3.870370 |
| 15 | 14 | 17 | 4.294118 |
| 16 | 15 | 16 | 3.250000 |
| 17 | 16 | 17 | 3.823529 |
| 18 | 17 | 103 | 4.145631 |
| 19 | 18 | 19 | 3.368421 |
| 20 | 19 | 21 | 3.809524 |
| 21 | 20 | 21 | 4.619048 |
| 22 | 21 | 33 | 3.424242 |
| 23 | 22 | 20 | 1.600000 |
| 24 | 23 | 15 | 3.533333 |
| 25 | 24 | 10 | 3.400000 |
| 26 | 25 | 18 | 3.833333 |
| 27 | 26 | 33 | 2.242424 |
| 28 | 27 | 10 | 3.400000 |
| 29 | 28 | 50 | 3.800000 |
| 30 | 29 | 16 | 3.375000 |
| 31 | 30 | 12 | 3.166667 |
| 32 | 31 | 12 | 4.583333 |
| 33 | 32 | 11 | 2.818182 |
| 34 | 33 | 14 | 4.785714 |
| 35 | 34 | 10 | 4.100000 |

Figure 4: DBSCAN Summary

# 3 Recommendation Engine Using Popularity Matching

We find that the restaurant with the highest average rating is Evanston Games and Cafe, with a perfect rating of 5.00. This is 1.25 points higher than the overall average rating of 3.75. We also find that the restaurant with the most reviews is Campagnola, with 48 reviews. This is 26 more reviews than the median number of reviews across all restaurants, which is just 22.

### 3.0.1 Simple Recommendation Engine

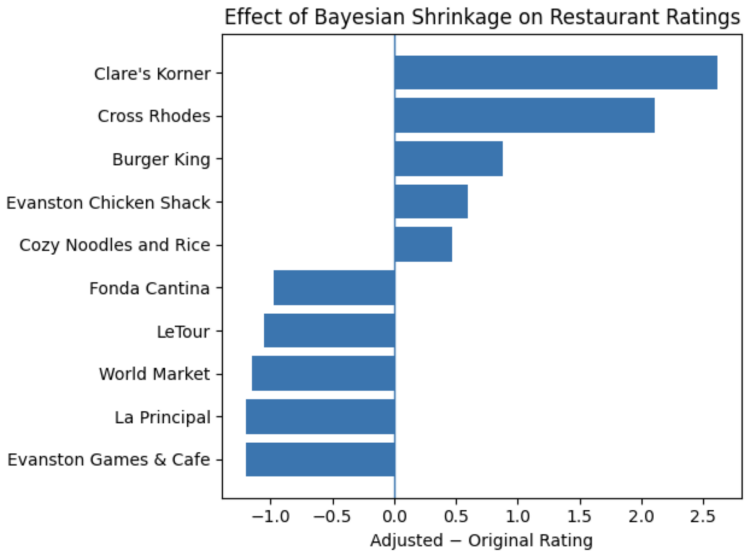
We first implement a simple recommendation engine that offers restaurant suggestions based on a specific cuisines entered by the user, based on the number of reviews the restaurant has. 5 shows us what this engine would recommend for Spanish, Chinese, Mexican cuisines, and Coffee.

| | Cuisine Type | Restaurant Name | num_reviews |
|---|---|---|---|
| 0 | Spanish | Tapas Barcelona | 29 |
| 1 | Spanish | 5411 Empanadas | 28 |
| 2 | Chinese | Peppercorns Kitchen | 33 |
| 3 | Chinese | Joy Yee Noodle | 31 |
| 4 | Chinese | Lao Sze Chuan | 24 |
| 5 | Mexican | Chipotle | 41 |
| 6 | Mexican | Taco Diablo | 37 |
| 7 | Mexican | Zentli | 17 |
| 8 | Mexican | Fridas | 13 |
| 9 | Mexican | Guzman y Gomez | 10 |
| 10 | Coffee | Pâtisserie Coralie | 30 |
| 11 | Coffee | Brothers K Coffeehouse | 15 |
| 12 | Coffee | Philz Coffee | 15 |
| 13 | Coffee | Evanston Games & Cafe | 1 |

Figure 5: Recommendations for Spanish, Chinese, Mexican food, and Coffee

### 3.0.2 Complex Recommendation Engine

We also implement a more complex recommendation engine that uses the Bayesian shrinkage estimator to score the popularity of each restaurant, instead of just the total number of reviews the restaurant has. We see in **??** that Clare's Korner had the greatest increase in its rating by using the Bayesian Shrinkage estimator, compared to its original rating. This is followed by Cross Rhodes, Burger King, Evanston Chicken Shack, and Cozy Noodles and Rice. We see that Evanston Games and Cafe, La Principal, World Market, LeTour, and Fonda Cantina were the most negatively impacted, on the other hand.



# 4 Content-based Filtering

## 4.1 Comparing Restaurants via Distance Metrics

Peppercorns Kitchen and Lao Sze Chuan show a very small Euclidean distance (0.2620) and a high cosine similarity (0.9842), indicating they are highly similar — consistent with both being Szechuan Chinese restaurants with similar attributes. In contrast, Peppercorns Kitchen and Epic Burger have a larger Euclidean distance (1.4287) and a moderate cosine similarity (0.5160), reflecting greater differences in cuisine type and other features. Overall, the metrics now appropriately capture meaningful distinctions between the restaurants.

## 4.2 Recommendations for Willie Jacobsen

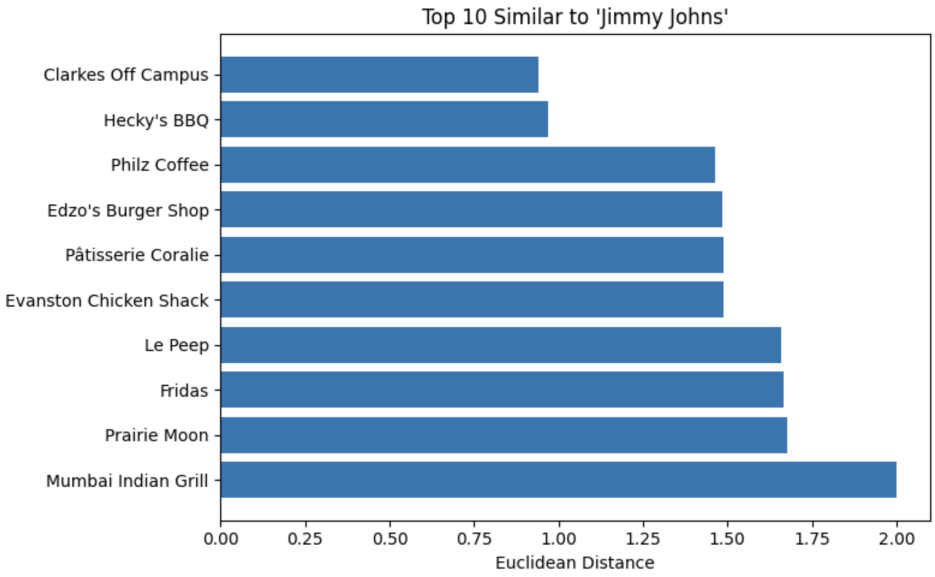



Figure 6: Willie Jacobsen's Top 10 Recommended Restaurants

As we see in 6, the most similar restaurant to Jimmy John's, which is Willie Jacobsen's favorite restaurant based on average ratings, is Mumbai Indian Grill. We can see the rest of the restaurants in order that would be recommended using our system in 6.

## 4.3 Comparing the Recommendations made by Euclidean vs Cosine Dist.

7 shows us the average of the mean ratings of the ten recommended restaurants, based on the input restaurant (x-axis) that each of the two recommendation systems give. We do this for twenty randomly selected input restaurants. We find that the results vary per restaurant - for some, the recommendation system that used Euclidean distance recommended better restaurants on average, while sometimes this was not the case. However, the recommendation system using cosine distance recommended restaurants that were, on average, 0.037 points lower in mean rating than those that were recommended by the system using Euclidean distance. This suggests that the latter recommendation system is slightly better, although this difference is negligible.

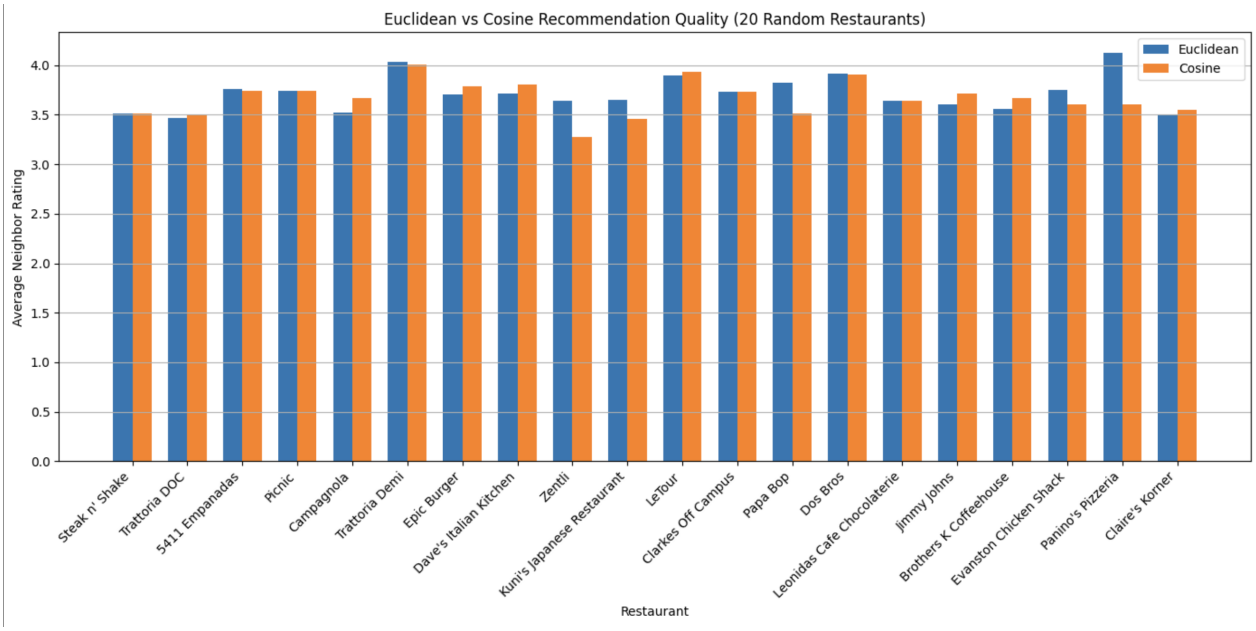



Figure 7: Bar Graph showing Average Ratings of Recommended Restaurants

# 5 Natural Language Analysis Version A

## 5.1 Jaccard Distances between Pairs of Restaurants

We see the following Jaccard distances between the selected restaurant pairs in table below: It makes sense that the

| **Restaurant Pair** | **Jaccard Distance** |
|---|---|
| Burger King vs. Edzo's Burger Shop | 0.7727 |
| Burger King vs. Oceanique | 0.8846 |
| Lao Sze Chuan vs. Kabul House | 0.8182 |

Table 1: Jaccard distances between selected restaurant pairs for Question 12

Jaccard distance between Burger King and Edzo's Burger Shop is smaller than that between the former and Oceanique, since the first pair are of the same cuisine. Furthermore, even though Burger King and Oceanique, and Lao Sze Chuan and Kabul House are both separate cuisines, we see that the latter pair has a smaller Jaccard distance. This may be due to other similarities such as the fact that they're both sit-in, and not fast-food restaurants. Burger King and Oceanique vary greatly because the former servers fast food and the latter is more fine dining.

## 5.2 Recommending Restaurants for Calvin Smith and Solomon M

| **User** | **Favorite Restaurant** | **Recommended Restaurant** | **Jaccard Distance** |
|---|---|---|---|
| Calvin Smith | Barn Steakhouse | Alcove | 0.7407 |
| Solomon M | Hokkaido Ramen | Tomo Japanese Street Food | 0.7826 |

Table 2: Favorite and recommended restaurants with Jaccard distances

# 6 Conclusion

In this project, we developed a variety of recommender systems for restaurants in Evanston, IL, using both structured metadata and natural language analysis. Through popularity matching, clustering, and content-based filtering approaches, we were able to suggest meaningful restaurant alternatives for users based on their preferences and review patterns. Distance metrics like Euclidean, cosine similarity, and Jaccard distance provided insights into restaurant similarities beyond just cuisine type, capturing factors like dining style and overall experience. Our results show that nuanced methods such as Bayesian shrinkage and natural language processing can significantly refine recommendation quality, ultimately helping users discover restaurants